Ming Fu (付明)

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Academic Curriculum Vitae | Last updated: July 2026

Research Profile

Researcher in operating systems and formal verification, with a focus on concurrency verification, weak memory consistency, OS architecture, program logics, and practical verification of systems software. My recent work bridges academic methods and deployed industrial systems, including verified memory management, scalable mobile graphics and rendering services, efficient tracing, and synchronization primitives for weak memory architectures. During my Huawei assignment in Dresden, I founded a new research center and led a team working on formal verification of concurrent programs, including the VSync/libvsync open-source verification effort.

Research interests: operating systems; multicore scalability; concurrency verification; weak memory models; formal verification of operating systems and processors; program logics; refinement verification; interactive theorem proving.

Appointments

Chief Expert, Huawei Fields Lab (华为菲尔兹实验室首席专家) **Jul. 2026–Present**
Huawei Technologies Co., Ltd. *Hangzhou, China*

- Serve as chief expert for industry-oriented research on OS architecture, multicore scalability, weak memory consistency, and formal verification of operating systems and processors.
- Provide technical leadership connecting formal methods, systems architecture, mobile OS services, rendering systems, synchronization, and Web-engine infrastructure.

Director, Huawei Fields Lab (华为菲尔兹实验室主任) **Aug. 2023–Jun. 2026**
Huawei Technologies Co., Ltd. *Hangzhou, China*

- Led industry-oriented research on OS architecture, multicore scalability, weak memory consistency, and formal verification of operating systems and processors.
- Built research programs that connect formal methods, systems architecture, mobile OS services, rendering systems, synchronization, and Web-engine infrastructure.

Founder and Director, Huawei Dresden Research Center **2019–2023**
Huawei Technologies Co., Ltd. *Dresden, Germany*

- Expatriated to Germany to establish Huawei Dresden Research Center from scratch and served as its founding director.
- Recruited and led a research team working on formal verification of concurrent programs, weak memory consistency, synchronization primitives, and systems software reliability.
- Led the VSync research line and the `open-s4c/libvsync` open-source project, a verified library of synchronization primitives and concurrent data structures.
- The VSync work received the ASPLOS 2021 Distinguished Paper Award.

Senior Expert, OS Kernel Lab **Jul. 2017–Mar. 2019**
Huawei Technologies Co., Ltd. *China*

- Served as a senior expert for the Hongmeng OS project.
- Led the formal verification team for key components of the Hongmeng microkernel and participated in microkernel design and development.

Associate Professor **Aug. 2016–Jun. 2017**
University of Science and Technology of China *Hefei, China*

- Applied an OS verification framework to the verification of SpaceOS, an operating system developed by a Chinese corporation.

Postdoctoral Researcher **Jan. 2011–Jul. 2016**
University of Science and Technology of China *Hefei, China*

- Developed concurrent program logics and refinement verification techniques for fine-grained concurrent programs.
- Applied refinement verification techniques to software transactional memory algorithms.
- Led a verification group of one Ph.D. student and five master's students to verify the commercial real-time embedded OS kernel μC/OS-II in Coq. Advisor: Xinyu Feng.

Visiting Assistant in Research **Nov. 2009–Oct. 2010**
Yale University *New Haven, CT, USA*

- Developed program logic for verifying optimistic concurrent programs. Advisor: Zhong Shao.

Education

Ph.D. in Computer Science **Jul. 2010**
University of Science and Technology of China *Hefei, China*

- Dissertation: Formal Verification of Concurrent Assembly Code (Chinese).
- Advisors: Yu Zhang and Yiyun Chen.

B.S. in Computer Science **Jul. 2004**
University of Science and Technology of China *Hefei, China*

Selected Research Systems and Open Source

- **VSync/libvsync**. Led the research and open-source development of `open-s4c/libvsync`, a verified library of synchronization primitives and concurrent data structures for weak memory architectures. The project grew out of the VSync research line at Huawei Dresden Research Center and is associated with the ASPLOS 2021 Distinguished Paper Award.

Publications

* *denotes corresponding author.*

1. Jiawei Wang, **Ming Fu***, Ruixian Wang, Chao Xu, Jonas Oberhauser, and Haibo Chen. "jwmalloc: A Verified Memory Allocator for Mobile Devices." *to appear in OSDI 2026, July 13–15, 2026.*
2. Yuanpei Wu, Dong Du, Chao Xu, Yubin Xia, Yang Yu, **Ming Fu**, Binyu Zang, and Haibo Chen. "OS Rendering Service Made Parallel with Out-of-Order Execution and In-Order Commit." *OSDI 2025, July 7–9, 2025.*
3. Yuanpei Wu, Dong Du, Chao Xu, Yubin Xia, **Ming Fu**, Binyu Zang, and Haibo Chen. "D-VSync: Decoupled Rendering and Displaying for Smartphone Graphics." *ASPLOS 2025, Mar. 2025.*
4. Jiawei Wang, Nian Liu, Arnau Casadevall-Saiz, Yutao Liu, Diogo Behrens, **Ming Fu**, Ning Jia, Hermann Haertig, and Haibo Chen. "Enabling Efficient Mobile Tracing with BTrace." *ASPLOS 2025, Mar. 2025.*
5. Jiawei Wang, Bohdan Trach, **Ming Fu***, Diogo Behrens, Jonathan Schwender, Yutao Liu, Jitang

Lei, Viktor Vafeiadis, Hermann Haertig, and Haibo Chen. "BWoS: Formally Verified Block-based Work Stealing for Parallel Processing." *OSDI 2023, July 10–12, 2023.*

6. Martin Beck, Koustubha Bhat, Lazar Stricevic, Geng Chen, Diogo Behrens, **Ming Fu***, Viktor Vafeiadis, Haibo Chen, and Hermann Haertig. "AtoMig: Automatically Migrating Millions Lines of Code from TSO to WMM." *ASPLOS 2023, Mar. 2023.*
7. Jiawei Wang, Diogo Behrens, **Ming Fu***, Lilith Oberhauser, Jonas Oberhauser, Jitang Lei, Geng Chen, Hermann Haertig, and Haibo Chen. "BBQ: A Block-based Bounded Queue for Exchanging Data and Profiling." *USENIX ATC 2022, July 11–13, 2022.*
8. Antonio Paolillo, Hernan Ponce-de-Leon, Thomas Haas, Diogo Behrens, Rafael Chehab, **Ming Fu**, and Roland Meyer. "Verifying and Optimizing Compact NUMA-Aware Locks on Weak Memory Models." *Technical Note, arXiv, July 9, 2022.*
9. Rafael Lourenco de Lima Chehab, Antonio Paolillo, Diogo Behrens, **Ming Fu***, Hermann Haertig, and Haibo Chen. "CLoF: A Compositional Lock Framework for Multi-level NUMA Systems." *SOSP 2021, Oct. 25–28, 2021.*
10. Jonas Oberhauser, Lilith Oberhauser, Antonio Paolillo, Diogo Behrens, **Ming Fu**, and Viktor Vafeiadis. "Verifying and Optimizing the HMCS Lock for Arm Servers." *NETYS 2021, May 2021.*
11. Jonas Oberhauser, Rafael Lourenco de Lima Chehab, Diogo Behrens, **Ming Fu***, Antonio Paolillo, Lilith Oberhauser, Koustubha Bhat, Yuzhong Wen, Haibo Chen, Jaeho Kim, and Viktor Vafeiadis. "VSync: Push-Button Verification and Optimization for Synchronization Primitives on Weak Memory Models." *ASPLOS 2021, Distinguished Paper Award, Apr. 2021.*
12. Jiawei Wang, **Ming Fu**, Lei Qiao, and Xinyu Feng. "Formalizing SPARCv8 Instruction Set Architecture in Coq." *Science of Computer Programming, 187:102371, 2020.*
13. Mo Zou, Haoran Ding, Dong Du, **Ming Fu**, Ronghui Gu, and Haibo Chen. "Using Concurrent Relational Logic with Helper for Verifying the AtomFS File System." *SOSP 2019, Oct. 27–30, 2019.*
14. Fengwei Xu, **Ming Fu***, Xinyu Feng, Xiaoran Zhang, Hui Zhang, and Zhaohui Li. "A Practical Verification Framework for Preemptive OS Kernels." *CAV 2016, pp. 59–79, July 2016.*
15. Jingyuan Cao, **Ming Fu***, and Xinyu Feng. "Practical Tactics for Verifying C Programs in Coq." *CPP 2015, pp. 97–108, Jan. 2015.*
16. Xiaoxiao Yang, Yu Zhang, **Ming Fu**, and Xinyu Feng. "A Temporal Programming Model with Atomic Blocks Based on Projection Temporal Logic." *Frontiers of Computer Science, 8(6):958–967, 2014.*
17. Hongjin Liang, Xinyu Feng, and **Ming Fu**. "Rely- Guarantee-Based Simulation for Compositional Verification of Concurrent Program Transformations." *ACM Transactions on Programming Languages and Systems, 36(1), Article 3, Mar. 2014.*
18. Yanni Kouskoulas, **Ming Fu**, Zhong Shao, and Peter Kazanzides. "Applying Mathematical Logic to Create Zero-Defect Software." *Johns Hopkins APL Technical Digest, 32(2), 2013.*
19. Xiaoxiao Yang, Yu Zhang, **Ming Fu**, and Xinyu Feng. "A Concurrent Temporal Programming Model with Atomic Blocks." *ICFEM 2012, pp. 22–37, Nov. 2012.*
20. Hongjin Liang, Xinyu Feng, and **Ming Fu**. "A Rely- Guarantee-Based Simulation for Verifying Concurrent Program Transformations." *POPL 2012, pp. 455–468, Jan. 2012.*
21. Zipeng Zhang, Xinyu Feng, **Ming Fu**, Zhong Shao, and Yong Li. "A Structural Approach to Prophecy Variables." *TAMC 2012, pp. 61–71, 2012.*
22. Yanni Kouskoulas, **Ming Fu**, Zhong Shao, and Peter Kazanzides. "Certifying the Concurrent State Table Implementation in a Surgical Robotic System." *Joint Workshop on High Confidence Medical Devices, Software, and Systems and Medical Device Plug-and-Play Interoperability, June 2011.*
23. **Ming Fu**, Yong Li, Xinyu Feng, Zhong Shao, and Yu Zhang. "Reasoning about Optimistic Concurrency Using a Program Logic for History." *CONCUR 2010, LNCS 6269, pp. 388–402, Aug. 2010.*

24. Yong Li, Yu Zhang, Yiyun Chen, and **Ming Fu**. "Formal Reasoning about Lazy-STM Programs." *Journal of Computer Science and Technology, 25(4):841–852, 2010.*
25. **Ming Fu**, Yu Zhang, and Yong Li. "Formal Verification of Concurrent Programs with Read-Write Locks." *Frontiers of Computer Science, 4(1):65–77, Jan. 2010.*
26. **Ming Fu**, Yu Zhang, and Yong Li. "Formal Reasoning about Concurrent Assembly Code with Reentrant Locks." *TASE 2009, pp. 233–240, July 2009.*
27. Yong Li, Yu Zhang, Yiyun Chen, and **Ming Fu**. "On the Verification of Strong Atomicity in Programs Using STM." *SSIRI 2009, pp. 117–125, July 2009.*
28. **Ming Fu** and Yu Zhang. "Homomorphism Resolving of XPath Trees Based on Automata." *APWeb/WAIM 2007, June 2007.*

Selected Invited Talks and Public Presentations

- Opening Remarks, Future Web. GOSIM Hangzhou 2025, Hangzhou, China, Sept. 13, 2025.
- Opening remarks and forum producer, Web Forum. OpenHarmony Technology Conference 2025, Changsha, China, Sept. 27, 2025.
- End-to-End Industrial Practice for High-Performance Safe Concurrency.
  端到端高性能安全并发的工业实践. CCF China Software Conference 2024, Formal Methods in Systems Software Forum, Xi'an, China, Nov. 17, 2024.
- Industrial practice of improving performance through verification.
  后摩尔时代通过验证提升性能的工业实践. CNCC 2022, Guiyang, China, Dec. 9, 2022.
- Taking Formal Verification of Systems Software from Academia to Industry. Verified Software Workshop Programme (VeTSS), UK, Sept. 24, 2019.
- Academic report, School of Computer Science and Technology, University of Science and Technology of China, Dec. 22, 2019.
- A Practical Verification Framework for Preemptive OS Kernels. National Software Analysis and Verification Workshop (SAVE 2016), Dec. 2016.
- Practical Tactics for Verifying C Programs in Coq. CPP 2015, Mumbai, India, Jan. 2015.
- A refinement-based verification framework for lock-based software transactional memory. SAVE 2014, Beijing, China, 2014.
- Reasoning about Optimistic Concurrency Using a Program Logic for History. CONCUR 2010; IBM Programming Language Day; Yale Programming Language Seminar, 2010.

Teaching

- Instructor, *Multicore Programming*, graduate-level course, School of Software, University of Science and Technology of China, 2012–2016.
- Instructor, *Frontier of Research on High-Confidence Software*, undergraduate-level course, University of Science and Technology of China, Summer 2012.

Professional Service and Community Leadership

- Chair, Formal Methods in Systems Software Forum, CCF China Software Conference 2024.
- Lead, OpenHarmony Concurrency and Collaboration TSG, as listed in public OpenHarmony TSC operations reports.
- Forum producer and opening speaker, Web Forum, OpenHarmony Technology Conference 2025.
- Forum producer, OS Kernel and Window Forum, OpenHarmony Technology Conference 2025.

- Journal reviewer: Journal of Software; Frontiers of Computer Science.
- Conference reviewer: LICS 2015; ITP 2015; ESOP 2013.

Grants

- Principal Investigator, *Verifying Lock-Free Concurrent Data Structures*, National Science Foundation of China, Grant No. 61103023, RMB 315,000, Jan. 2012–Dec. 2014.
- Principal Investigator, *Refinement-Based Verification Framework for Software Transactional Memory*, Fundamental Research Funds for the Central Universities, Grant No. WK0110000031, RMB 75,000, Jan. 2012–Dec. 2013.
- Principal Investigator, China Postdoctoral Science Foundation, Grant No. 2012M511420, RMB 50,000, Aug. 2012–Aug. 2013.

Honors and Awards

- Distinguished Paper Award, ASPLOS 2021, for *VSync: Push-Button Verification and Optimization for Synchronization Primitives on Weak Memory Models*.
- Student Fellowship for attending CONCUR 2010.
- China Scholarship Council Fellowship for visiting Yale University, 2009–2010.
- Third Prize Fellowship, University of Science and Technology of China, 2000.
- New Student Fellowship, University of Science and Technology of China, 1999.

Technical Expertise

- Operating-system architecture, multicore programming, weak memory consistency, and synchronization.
- Separation logic, concurrent program logic, and refinement-based program logic.
- Interactive theorem proving with Coq.
- Programming languages and tools: C/C++, Java, OCaml, Coq, and LaTeX.